Hospital Practice

MANAGEMENT OF THE PREGNANT DIABETIC: HOME OR HOSPITAL, WITH OR WITHOUT GLUCOSE METERS?

S. M. STUBBS J. M. BRUDENELL
D. A. PYKE P. J. WATKINS

Departments of Obstetrics and Gynæcology and Diabetes, King's College Hospital, London

W. A. STUBBS

Department of Medicine, St. Bartholomew's Hospital, London

K. G. M. M. ALBERTI

Department of Clinical Biochemistry and Metabolic Medicine, Royal Victoria Infirmary, Newcastle-upon-Tyne

Summary Thirteen pregnant insulin-dependent diabetic patients were allocated to blood-glucose self-monitoring (meter) or conventionally treated (non-meter) groups at 30–31 weeks. A day-profile of blood-glucose and intermediary metabolite levels was obtained 2 weeks later at home and after routine admission to hospital at 35–36 weeks. Metabolic profiles were also obtained in eight normal pregnant women of equivalent gestational ages admitted to hospital for study. The mean blood-glucose and metabolite concentrations in both the meter and non-meter groups were similar to those obtained in the non-diabetic pregnant women (mean diurnal blood-glucose 4·6±1·1 mmol/l, meter group; 5·3±1·5 non-meter group; 4·8±0·8 controls). The diabetic control achieved at home was not improved by the use of a meter or admission to hospital (mean glucose at home 5·0±1·3 and 6·3±1·6 mmol/l in hospital). Both the use of a meter and admission to hospital may be valuable in some patients. Neither, however, is essential for the good control of blood-glucose in all pregnant diabetics.

INTRODUCTION

GOOD control of maternal diabetes is an important determinant of a successful pregnancy. Such control has been achieved at King's College Hospital by clinic visits every 2–4 weeks and admission to hospital from the 32nd week of pregnancy. This policy has undoubtedly contributed to the fall in diabetic perinatal mortality during the past 20 years.

Recently there has been considerable interest in self-monitoring of blood-glucose by diabetics at home,[1-3] and improved diabetic control in pregnancy is said to be achieved by these means. We have, therefore, compared blood-glucose and metabolic control in patients treated conventionally with that in those who monitor their blood-glucose by the use of a meter. At the same time we have re-evaluated the effects of early hospital admission on diabetic control.

METHOD

Controls

Eight normal women, aged 18–30 at 32–36 weeks' gestation, were admitted to hospital for 24 h. A cannula was inserted into an antecubital vein and free-flowing hourly blood-samples were taken throughout the day for the analysis of blood glucose, lactate, alanine, glycerol, and 3-hydroxybutyrate by automated fluorimetric techniques.[4]

Diabetics

For at least 2 weeks before the study the seven patients randomly allocated to the meter group measured blood-glucose at home with 'Dextrostix' (Ames), using an 'Eyetone' meter. Their ages ranged from 23–31 years (mean 26±3), and they had had diabetes for 7–19 years (mean 13±4). Samples were taken 7 times a day twice each week (before and after each main meal and before bedtime). Insulin dosage and diet were adjusted by telephone or clinic consultation with one of us.

The six patients in the non-meter group checked their urine glucose 4 times daily, and their random blood-glucose levels were measured at the fortnightly clinic visits. Their ages ranged from 25–32 years (29±3), and they had had diabetes for 2–20 years (mean 11±6).

At 32–33 weeks' gestation all patients were visited at home (by S. M. S.), and a blood-glucose and metabolite profile was obtained for the period from before their normal morning insulin injection until bedtime. Samples were taken hourly from an indwelling antecubital venous cannula whilst patients did their normal domestic chores.

Blood-glucose and metabolite profiles were repeated in the same patients at 35–36 weeks' gestation, at least 10 days after their routine admission to hospital, where the patients were given their prescribed diet, visited daily by a member of the diabetic team, and encouraged to remain active.

Statistics

Results are expressed as mean ±SD in relation to the main meals of the day, since the time of insulin injection and meals varied for each patient at home. Comparisons have been made by the use of the Student's paired and unpaired *t* tests.

RESULTS

All patients were subsequently delivered of normal healthy babies, whose birth-weights ranged from 2·70 to 4·54 kg (mean 3·27±0·61) in the meter group, and from 2·80 to 4·14 kg (mean 3·41±0·60) in the non-meter group. The babies of the non-diabetics were all within 13% of their ideal body-weight. The baby of one patient in the non-meter group died at home at the age of 4 weeks. No abnormalities were found at necropsy, and cot death was diagnosed.

Comparison of Metabolic Profiles in Self-monitoring and Conventionally Treated Groups at Home

The mean concentration of blood glucose, lactate, alanine, and glycerol were similar throughout the day in meter, non-meter, and control groups (table I, fig. 1). The diurnal 3-hydroxybutyrate in the non-meter group was raised compared with controls, but not when compared with the meter group. Although there was no significant difference between the mean diurnal blood-glucose in the meter group (4·6±1·1 mmol/l), non-meter group (5·3±1·5 mmol/l), and controls (4·8±0·8 mmol/l), there were minor differences between the groups at certain sampling times. Thus, blood-glucose in

CONVERSION FACTORS
Blood-glucose, 1 mmol/l = 18·02 mg/dl
Blood-lactate, 1 mmol/l = 9·02 mg/dl

TABLE I—BLOOD-METABOLITE LEVELS IN PREGNANT DIABETICS AND IN NORMAL PREGNANT WOMEN

Group	Lactate (mmol/l)	Alanine (μmol/l)	Glycerol (μmol/l)	3-hydroxy-butyrate (μmol/l)
Diabetic				
Meter group	0·8±0·1	264±40	67±29	112±83
Non-meter group	0·8±0·2	277±73	71±11	105±42*
Controls	0·9±0·3	303±76	63±44	47±26

Results given are means ±SD of 14 samples taken through the day.
*p<0·05 compared with controls.

TABLE II—BLOOD-METABOLITE LEVELS OF PREGNANT DIABETICS AT HOME AND IN HOSPITAL

	Lactate (mmol/l)	Alanine (μmol/l)	Glycerol (μmol/l)	3-hydroxy-butyrate (μmol/l)
Home	0·8±0·1	270±55	69±22*	109±65†
Hospital	0·9±0·2	317±58	51±17	95±60†

Results given are means ±SD of 14 samples taken through the day.
*p<0·05 between home and hospital metabolites in diabetics.
†p <0·05 between home or hospital groups and controls.

the non-meter group was lower than that in the meter group after dinner, and blood-lactate was higher 4 h after lunch. Blood lactate and alanine were lower and blood-3-hydroxybutyrate higher after lunch in diabetics in the non-meter group compared with controls.

Comparison of Metabolic Profiles at Home and in Hospital

There were again no major significant differences in the mean blood glucose, lactate, or alanine throughout the day between diabetics when they were at home or in hospital and those in normal pregnant women (table II, (fig. 2). The diurnal blood-glucose was lower at home in nine of the thirteen patients, but the mean levels were similar (5·0±1·3 mmol/l at home and 6·3±1·6 mmol/l in hospital). Blood-glucose in hospital was higher than that of the controls (4·8±0·8 mmol/l, p<0·02), and the mean 3-hydroxybutyrate was lower in controls than in diabetics at home or in hospital (p<0·05). The circulating blood-glycerol was 35% higher when the patients were at home, but this value was not different from that for normal women.

Blood-levels at certain sampling times showed differences between the groups. Thus blood-glucose was higher in hospital in the diabetics during the morning and 2 h after dinner, compared with that in patients at home. There were also small differences in blood alanine and lactate at certain times of the day when values for the patients at home and in hospital were compared. The blood-glucose in hospital during the morning and after lunch was higher than that in normal pregnant women, whereas at home lower than normal values were found before lunch and after dinner.

DISCUSSION

In this study good control of blood-glucose and intermediary metabolites in pregnant diabetics was achieved at home by conventional methods; self-monitoring of blood-glucose with a meter proved no better.

Patients at home maintained somewhat better blood-glucose control than in hospital. The reasons for this are uncertain. The patients were 3–4 weeks later in gestation at the time of study in hospital, but there is no good evidence that the control of blood-glucose becomes more difficult late in pregnancy. Exercise can exert a major influence on blood-glucose control and, although patients were encouraged to be active in hospital, they were probably less active than they had been at home. At home patients frequently made small alterations in diet, time and dose of insulin, or exercise to maintain optimum regulation of their diabetes. The relative rigidity of hospital life may prevent this degree of flexibility.

Despite the relatively good blood-glucose control, there were still some differences in circulating blood-metabolites between diabetics and normal subjects. The significance of these differences is unknown. They may be significant since we have found that changes in maternal metabolism are associated with effects on fetal

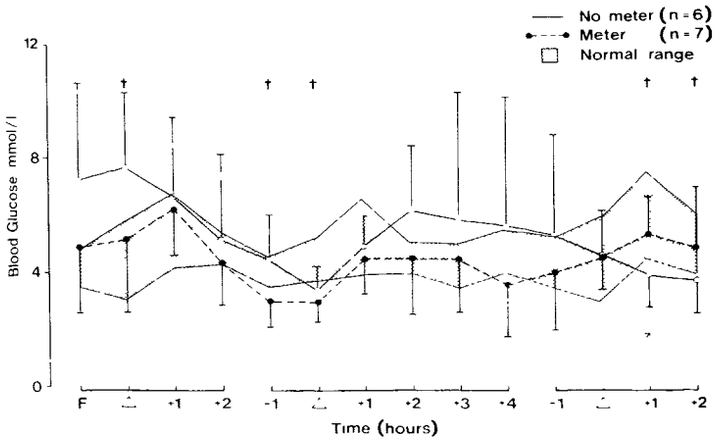

Fig. 1—Blood-glucose (mean±SD) in pregnant diabetics with or without a glucose meter.

* p<0·05 between meter and non-meter groups.
† p<0·05 between meter or non-meter groups and controls.

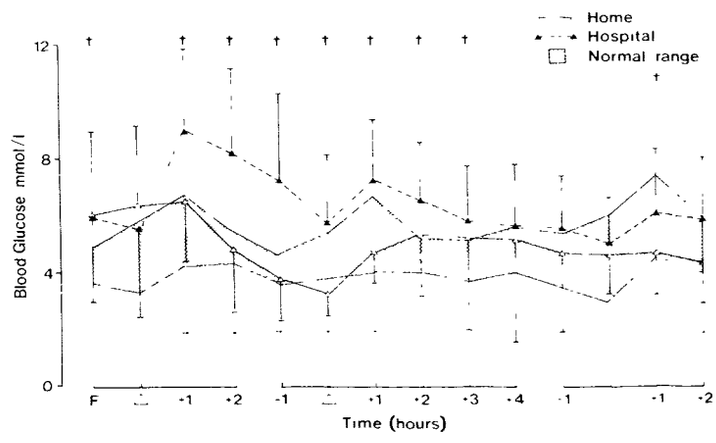

Fig. 2—Blood-glucose (mean±SD) in 13 pregnant diabetics at home and in hospital.

* p<0·05 between home and hospital groups.
† p<0·05 between home or hospital groups and controls.

behaviour as assessed by fetal movements measured by real-time ultrasound (Stubbs SM, unpublished).

We are re-examining our policy of hospital admission for all patients at 32 weeks and our management of the pregnant diabetic is now more flexible. Early admission is still recommended, where social or medical circumstances are unsatisfactory. Glucose self-monitoring has a role in patient management especially in early pregnancy in those patients in whom control is inadequate; it can also be a useful part of patient education, especially in pregnancy where it may help to motivate the patient to achieve the stricter control needed in pregnancy. However, conventional methods alone can be just as effective in some patients.

We thank J. Burrin, D. Cooper, S. Campbell, and the staff in the diabetic clinic for their help. The work was supported by a Medical Research Council Project Grant and the British Diabetic Association.

Requests for reprints should be addressed to S. M. S., Department of Obstetrics and Gynæcology, King's College Hospital, Denmark Hill, London SE5.

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Occasional Survey

TIME OF FERTILISATION AND SEX OF INFANTS

WILLIAM H. JAMES

Galton Laboratory, University College, London WC1

Summary The sex of a human zygote may depend on the time during the mother's menstrual cycle that it is formed. Direct data to support this hypothesis suggest that male zygotes tend to be formed both early and late in the fertile period, and female zygotes in the middle. Criticisms of the direct data seem to be only that measurements of the timing of events around conception are inexact. There are several kinds of indirect data: (1) the variation of sex ratio with war and with duration of marriage, and (2) the non-binomial distribution of the combination of the sexes in human dizygotic twins and other mammalian polyzygotic litters. The sex of a zygote may be determined by maternal gonadotrophin levels when it is formed. This elaboration of the hypothesis could explain the low sex ratio in Negroes.

INTRODUCTION

IT has often been suggested that the sex of a human zygote depends on the time during the menstrual cycle that it is formed—a hypothesis that has been put forward again by Guerrero[1] and Harlap.[2] Evidence bearing on this hypothesis is reviewed here. There are two sorts of data, direct and indirect.

DIRECT DATA

Guerrero's data on natural inseminations came from women who had failed with a rhythm method of contraception. Harlap's data were on births to Jewish women who had ritually abstained from coitus during the week after cessation of menses. Both authors reported more boys from inseminations early and late in the fertile period, and more girls from inseminations in the middle of the fertile period.

It is unlikely that these results were due to chance: however, some artifact might possibly be responsible rather than the phenomenon claimed.

Both these papers have elicited comment.[3,4] The women in Guerrero's study were said to be unrepresentative—they were predominantly Catholic women whose attempts to use the rhythm method had failed. These criticisms do not seem to have much bite since the critics do not explain *how* these features might cause the effect described by Guerrero. Harlap's paper has been criticised because mothers were interviewed about their cycle lengths on the first day postpartum when their responses might be even more unreliable than usual. She also used the formula Y=N−14, where Y is the expected ovulation day and N is the reported mean cycle length for each woman, which is only a rough approximation. Finally she assumed that Orthodox women have intercourse on the day of the ritual bath and that the intercourse is fertile. The substance of these criticisms is that the cycle day of fruitful insemination (relative to ovulation day) was inaccurately measured. However, if there was no bias in these measurements (but rather, random error) these comments, far from being criticisms, may be construed as support. The onus is on the critics to show that there is bias which is responsible for the effect.

Data which relate time of insemination and the sex of other mammalian offspring have been reviewed briefly.[1] Guerrero considered six studies dealing with natural insemination: in each there was a non-significant trend supporting the hypothesis. In view of the difficulty of interpreting the data on man, and of the equivocal support from data on other mammals, it is necessary to review indirect data.

Sex Ratio and Spontaneous Abortion

Before this, however, we should consider the view that variations in the secondary sex ratio are due to different spontaneous abortion rates of male and female fetuses (and that therefore the secondary sex ratio reflects no useful information on the primary sex ratio). This argument usually depends on two premises: firstly, that spontaneous abortions contain a predominance of male fetuses and, secondly, that spontaneous abortion rates are increased by adverse environmental circumstances (e.g., high maternal age, low social class).